

AKLT Model Analysis

This notebook analyzes the AKLT (Affleck-Kennedy-Lieb-Tasaki) model using matrix product states and explores its symmetry properties. Below is the code which finds the symmetry group's underlying Lie algebra's generator in both the physical level and the virtual level.

Import Required Libraries

```
=====
MODULE VALIDATION CHECK
=====

[1] Checking Centraliser.TensorProductGroup...
    ✓ TensorProductGroup class exists with required methods

[2] Checking Centraliser.TensorProductGroupSO (SO(N) framework)...
    ✓ TensorProductGroupSO class exists with required methods

[3] Checking Centraliser.SON (SO(N) basis)...
    ✓ SON class exists with so_basis method

[4] Checking find_generators_coefficients parameters...
    ✓ Parameters: ['self', 'M', 'verbose', 'basis_generators', 'reference_coeffs']

[5] Checking MatrixProductState.find_generators_coefficients wrapper...
    ✓ Wrapper method exists with parameters: ['self', 'M', 'basis', 'verbose', 'dim', 'reference_coeffs']
```

Module Validation

Initialize AKLT MPS and Parent Hamiltonian

The AKLT state was initialised with the built-in `aklt()` function of the `emachine_v2.py` script. The chain was also generated with a boundary matrix `np.array([[1,0],[0,1]])`, which is under the PBC condition. The parent hamiltonian is found using orthogonal construction of the MPS. The hamiltonian in consideration have local projectors of length 2, and total length 3.

Ground State Computation

$SU(N)$ Symmetry Analysis

The symmetry of the AKLT model was assumed to be continuous and belongs to some group isomorphic to the $SU(3)$ group on the physical level. A basis of the $\mathfrak{su}(3)$ Lie algebra generators are found using a construction of the Gell-Mann Basis. As the symmetry group G of the MPS is a group which commutes with the Parent Hamiltonian, we can construct the symmetry group by constructing the linear operator of the map $F := [-, H] : \mathfrak{g} \rightarrow \mathcal{M}^{d^l \times d^l}$. The symmetry group is then generated by the kernel of F .

===== SYMMETRIC GENERATORS ANALYSIS =====

Found 3 generator(s) in the null space:

Generator 1:

Null space basis vector index: 0

Coefficients in local su(3) basis (orthonormalized): [-0.7068 0. -0.7068 0.01
04 0. 0.0104 0.0115 0.02]

Basis generators in su(3) (with non-zero coefficients):

[0] (coeff: -0.7068):
[[0.+0.j 1.+0.j 0.+0.j]
[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: -0.7068):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: 0.0104):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: 0.0104):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: 0.0115):
[[0.7071+0.j 0. +0.j 0. +0.j]
[0. +0.j -0.7071+0.j 0. +0.j]
[0. +0.j 0. +0.j 0. +0.j]]
[7] (coeff: 0.02):
[[0.4082+0.j 0. +0.j 0. +0.j]
[0. +0.j 0.4082+0.j 0. +0.j]
[0. +0.j 0. +0.j -0.8165+0.j]]

Local su(3) generator (Hermitian enforced):

[[0.0163+0.j -0.7068-0.0104j 0. +0.j]
[-0.7068+0.0104j 0. +0.j -0.7068-0.0104j]
[0. +0.j -0.7068+0.0104j -0.0163+0.j]]

Generator 2:

Null space basis vector index: 1

Coefficients in local su(3) basis (orthonormalized): [-0.0098 0. -0.0098 -0.70
67 0. -0.7067 0.0163 0.0283]

Basis generators in su(3) (with non-zero coefficients):

[0] (coeff: -0.0098):
[[0.+0.j 1.+0.j 0.+0.j]
[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: -0.0098):
[[0.+0.j 0.+0.j 0.+0.j]

```

[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: -0.7067):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: -0.7067):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: 0.0163):
[[ 0.7071+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j -0.7071+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j 0.      +0.j]]
[7] (coeff: 0.0283):
[[ 0.4082+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j 0.4082+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j -0.8165+0.j]]

```

```

Local su(3) generator (Hermitian enforced):
[[ 0.0231+0.j      -0.0098+0.7067j 0.      +0.j      ]
[-0.0098-0.7067j 0.      +0.j      -0.0098+0.7067j]
[ 0.      +0.j      -0.0098-0.7067j -0.0231+0.j      ]]

```

Generator 3:

```

Null space basis vector index: 2
Coefficients in local su(3) basis (orthonormalized): [-0.0166  0.      -0.0166 -0.02
28 0.      -0.0228 -0.4996 -0.8653]

```

```

Basis generators in su(3) (with non-zero coefficients):
[0] (coeff: -0.0166):
[[0.+0.j 1.+0.j 0.+0.j]
[1.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[2] (coeff: -0.0166):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 1.+0.j]
[0.+0.j 1.+0.j 0.+0.j]]
[3] (coeff: -0.0228):
[[0.+0.j 0.-1.j 0.+0.j]
[0.+1.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.+0.j]]
[5] (coeff: -0.0228):
[[0.+0.j 0.+0.j 0.+0.j]
[0.+0.j 0.+0.j 0.-1.j]
[0.+0.j 0.+1.j 0.+0.j]]
[6] (coeff: -0.4996):
[[ 0.7071+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j -0.7071+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j 0.      +0.j]]
[7] (coeff: -0.8653):
[[ 0.4082+0.j 0.      +0.j 0.      +0.j]
[ 0.      +0.j 0.4082+0.j 0.      +0.j]
[ 0.      +0.j 0.      +0.j -0.8165+0.j]]

```

```

Local su(3) generator (Hermitian enforced):

```

```

[[-0.7065+0.j      -0.0166+0.0228j  0.      +0.j      ]
 [-0.0166-0.0228j  0.      +0.j      -0.0166+0.0228j]
 [ 0.      +0.j      -0.0166-0.0228j  0.7065+0.j      ]]

```

Orthonormalized null space basis (3 vectors):

Basis vector 1 (norm = 1.0000):

```
[-0.7068  0.      -0.7068  0.0104  0.      0.0104  0.0115  0.02  ]
```

Basis vector 2 (norm = 1.0000):

```
[-0.0098  0.      -0.0098 -0.7067  0.      -0.7067  0.0163  0.0283]
```

Basis vector 3 (norm = 1.0000):

```
[-0.0166  0.      -0.0166 -0.0228  0.      -0.0228 -0.4996 -0.8653]
```

We see that the generators span the $\mathfrak{so}(3)$ Lie algebra in the spin-1 representation, which is consistent with that in the literature.

Interpretation of the Physical-Space Output

The output above provides generators T_1, T_2, T_3 from the physical-space symmetry analysis of the AKLT parent Hamiltonian. These are elements of a Lie algebra obtained from the kernel of the commutator map $F(X) = [X, H]$, so they correspond to transformations that preserve the Hamiltonian symmetry.

Applying the combined generator $T_1 + 2T_2$ with angle $\theta = \pi/2$ produces a transformed MPS `umps`, and the state `ustate` is its corresponding transformed ground-space state. This demonstrates that the symmetry action maps the AKLT ground space to another valid state within the same symmetry-protected manifold.

Virtual Symmetry Analysis

After obtaining the physical symmetry group generators $\mathfrak{g}$, the corresponding virtual symmetry group generators $\mathfrak{v}$ can be obtained from the relation: $U^{xy} A_{ij}^y = e^{i\epsilon\phi} V_{ik} A_{kl}^x V_{lj}^{-1}$, expanding $U^{xy} = e^{i\epsilon T^{xy}}$ and $V_{ij} = e^{i\epsilon K_{ij}}$, to first order of ϵ , the expression becomes $T^{xy} A_{ij}^y = [K, A^x]_{ij} + \phi A_{ij}^x$. We get a set of overdetermined linear equations, hence the generators are obtained by numerical optimisation.

```
Physical generator G:
[[ 0.0163+0.j      -0.7068-0.0104j  0.      +0.j      ]
 [-0.7068+0.0104j  0.      +0.j      -0.7068-0.0104j]
 [ 0.      +0.j      -0.7068+0.0104j -0.0163+0.j      ]]
```

```
Virtual generator found:
Phase: -0.000000+0.000000j
Error: 1.782291e-15
Relative error: 1.543715e-15
Generator K:
[[-0.0082+0.j      0.4998-0.0073j]
 [ 0.4998+0.0073j  0.0082+0.j      ]]
```

```
Physical generator G:
[[ 0.0231+0.j      -0.0098+0.7067j  0.      +0.j      ]
 [-0.0098-0.7067j  0.      +0.j      -0.0098+0.7067j]
 [ 0.      +0.j      -0.0098-0.7067j -0.0231+0.j      ]]
```

```
Virtual generator found:
Phase: -0.000000-0.000000j
Error: 8.068936e-15
Relative error: 6.989764e-15
Generator K:
[[-0.0115+0.j      0.0069+0.4997j]
 [ 0.0069-0.4997j  0.0115+0.j      ]]
```

```
Physical generator G:
[[-0.7065+0.j      -0.0166+0.0228j  0.      +0.j      ]
 [-0.0166-0.0228j  0.      +0.j      -0.0166+0.0228j]
 [ 0.      +0.j      -0.0166-0.0228j  0.7065+0.j      ]]
```

```
Virtual generator found:
Phase: -0.000000-0.000000j
Error: 7.395017e-16
Relative error: 9.049783e-16
Generator K:
[[ 0.3533+0.j      0.0118+0.0162j]
 [ 0.0118-0.0162j -0.3533+0.j      ]]
```

Successfully extracted K1, K2, K3 from virtual_symmetry output.

Interpretation of the Virtual-Space Output

The printed matrices V_1, V_2, V_3 are virtual (bond-space) symmetry generators extracted from the same parent Hamiltonian. They represent the symmetry action on the internal MPS degrees of freedom and should satisfy the same algebraic structure as the physical generators.

The next cell checks the Jacobi identity for (V_1, V_2, V_3) ; a vanishing or simplified result confirms that these generators form a consistent Lie algebra representation on the virtual space.

Jacobi Identity Verification

To verify that the virtual generators indeed form a valid Lie Algebra, apply the Jacobi Identity to the three generators.

```
[[0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j]]
```

Finding 2-cocycle of AKLT Model

We want to get the relation between virtual symmetry group elements. As V is a projective representation of G , elements in V satisfy some relation: $V_g V_h = \omega(g, h) V_{gh}$, where $\omega : G \times G \rightarrow U(1)$, also know as the 2-cocycle of the group G . Here we want to calculate the form of ω by considering different elements from G .

```
Group element g:  $\theta = [0.7853981633974483, 0, 0]$ 
Group element h:  $\theta = [0, 0.7853981633974483, 0]$ 
2-cocycle  $\omega(g,h) = (1-1.6471212663260437e-17j)$ 
Phase of  $\omega(g,h) = -0.000000$  rad
Verification error:  $6.60e-16$ 
```

Computing the 2-cocycle for Multiple Group Elements

Now let's compute $\omega(g,h)$ for various combinations to see the structure of the 2-cocycle.

2-cocycle $\omega(g,h)$ for various group elements:

g angles	h angles	$\omega(g,h)$ phase	Error
[0, 0, 0]	[0, 0, 0]	0.0000	0.00e+00
[0, 0, 0]	[0.5235987755982988, 0, 0]	0.0000	3.71e-16
[0, 0, 0]	[0.7853981633974483, 0, 0]	0.0000	2.99e-16
[0, 0, 0]	[1.0471975511965976, 0, 0]	0.0000	9.88e-16
[0, 0, 0]	[1.5707963267948966, 0, 0]	0.0000	6.41e-16
[0, 0, 0]	[3.141592653589793, 0, 0]	-0.0000	1.04e-15
[0, 0, 0]	[6.283185307179586, 0, 0]	-3.1416	1.29e-15
[0.5235987755982988, 0, 0]	[0, 0, 0]	0.0000	3.71e-16
[0.5235987755982988, 0, 0]	[0.5235987755982988, 0, 0]	-0.0000	2.32e-16
[0.5235987755982988, 0, 0]	[0.7853981633974483, 0, 0]	-0.0000	6.06e-16
[0.5235987755982988, 0, 0]	[1.0471975511965976, 0, 0]	0.0000	4.61e-16
[0.5235987755982988, 0, 0]	[1.5707963267948966, 0, 0]	0.0000	5.59e-16
[0.5235987755982988, 0, 0]	[3.141592653589793, 0, 0]	-3.1416	1.97e-15
[0.5235987755982988, 0, 0]	[6.283185307179586, 0, 0]	-3.1416	1.16e-15
[0.7853981633974483, 0, 0]	[0, 0, 0]	0.0000	2.99e-16
[0.7853981633974483, 0, 0]	[0.5235987755982988, 0, 0]	-0.0000	7.93e-16
[0.7853981633974483, 0, 0]	[0.7853981633974483, 0, 0]	-0.0000	1.36e-15
[0.7853981633974483, 0, 0]	[1.0471975511965976, 0, 0]	-0.0000	1.27e-15
[0.7853981633974483, 0, 0]	[1.5707963267948966, 0, 0]	-0.0000	1.06e-15
[0.7853981633974483, 0, 0]	[3.141592653589793, 0, 0]	-3.1416	3.48e-15
[0.7853981633974483, 0, 0]	[6.283185307179586, 0, 0]	-3.1416	1.47e-15

All verification errors < $1e-10$: True

```
/opt/anaconda3/envs/physics_3_11_9/lib/python3.11/site-packages/numpy/linalg/_linalg.py:2406: RuntimeWarning: divide by zero encountered in det
  r = _umath_linalg.det(a, signature=signature)
/opt/anaconda3/envs/physics_3_11_9/lib/python3.11/site-packages/numpy/linalg/_linalg.py:2406: RuntimeWarning: invalid value encountered in det
  r = _umath_linalg.det(a, signature=signature)
```

Summary and Interpretation of 2-Cocycle Results

The 2-cocycle $\omega(g, h)$ characterizes the **projective representation** structure of the virtual symmetry group. Key observations:

What the 2-cocycle tells us:

1. Projective vs. Linear Representation:

- If $\omega(g, h) = 1$ for all g, h , the virtual representation is **linear**: $V_g V_h = V_{gh}$
- If $\omega(g, h) \neq 1$ for some g, h , the representation is **projective**: $V_g V_h = \omega(g, h) V_{gh}$
- The AKLT model typically exhibits non-trivial ω , indicating a projective representation on the virtual space

2. Phase Structure:

- The phase $\arg(\omega(g, h))$ reveals how the symmetry "twists" when composed in the virtual space
- For commuting group elements ($g \cdot h = h \cdot g$), we expect $\omega(g, h) = \omega(h, g)$
- The distribution of phases can indicate the cohomology class of the projective representation

3. Symmetry Protection:

- Non-trivial 2-cocycles are signatures of **symmetry-protected topological (SPT) phases**
- The AKLT state is a canonical example of a 1D SPT phase with $SO(3)$ symmetry
- The projective virtual representation indicates edge states cannot be removed without breaking symmetry

4. Verification Errors:

- Small errors ($< 10^{-10}$) confirm the relation $V_g V_h = \omega(g, h) V_{gh}$ holds numerically
- Larger errors might indicate:
 - Generator coefficients from different matrix logarithm branches
 - Numerical instability in matrix exponentiation
 - The generators don't form a closed Lie algebra

Physical Significance:

The projective nature of the virtual representation is **essential** to the AKLT model's topological properties:

- The virtual Hilbert space has dimension 2 (spin-1/2), while physical space has dimension 3 (spin-1)
- The spin-1/2 edge states provide a **fractionalization** of the bulk spin-1 symmetry
- This fractionalization is guaranteed by the non-trivial projective representation structure

If all $\omega(g, h) \approx 1$, the state would be in a **trivial phase** where edge modes can be removed by local operations.

Cohomology and 2-Cocycles — Structured Summary

1. Projective Representation

Virtual symmetry operators satisfy:

$$V_g V_h = \omega(g, h) V_{gh}$$

where:

$$\omega(g, h) \in U(1)$$

The phase function ω is the central object.

2. 2-Cocycle Condition

Associativity requires:

$$(V_g V_h) V_k = V_g (V_h V_k)$$

This implies:

$$\omega(g, h) \omega(gh, k) = \omega(h, k) \omega(g, hk)$$

A function satisfying this condition is called a **2-cocycle**.

Denote the set of all such functions:

$$Z^2(G, U(1))$$

3. Gauge Freedom (2-Coboundary)

Virtual operators are not unique:

$$V_g \rightarrow e^{i\beta(g)} V_g$$

with arbitrary function:

$$\beta : G \rightarrow \mathbb{R}$$

Under this transformation:

$$\omega(g, h) \rightarrow \omega(g, h) \frac{e^{i\beta(g)} e^{i\beta(h)}}{e^{i\beta(gh)}}$$

The extra factor is called a **2-coboundary**.

Denote all such coboundaries:

$$B^2(G, U(1))$$

4. Cohomology Group Definition

Two cocycles differing by a coboundary describe the same physics.

Define the second cohomology group:

$$H^2(G, U(1)) = Z^2(G, U(1)) / B^2(G, U(1))$$

Interpretation:

Distinct projective symmetry classes.

5. Compact Lie Group Shortcut

For compact connected Lie groups:

$$H^2(G, U(1)) \cong \text{Hom}(\pi_1(G), U(1))$$

So classification reduces to computing the fundamental group:

$$\pi_1(G)$$

6. Example: SO(3)

Topology:

$$SO(3) = SU(2) / \mathbb{Z}_2$$

Fundamental group:

$$\pi_1(SO(3)) = \mathbb{Z}_2$$

Homomorphisms:

$$\text{Hom}(\mathbb{Z}_2, U(1)) = \{+1, -1\}$$

Therefore:

$$H^2(SO(3), U(1)) = \mathbb{Z}_2$$

7. Physical Meaning

Two classes:

1. Trivial class:

$$V_{2\pi} = +1$$

(Integer spin)

2. Non-trivial class:

$$V_{2\pi} = -1$$

(Half-integer spin)

This minus sign cannot be removed by gauge.

8. Geometric Origin

A 2π rotation in $SO(3)$:

- Is a closed loop
- Lifts in $SU(2)$ from $+1$ to -1
- Is non-contractible

Thus:

$$\pi_1(SO(3)) = \mathbb{Z}_2$$

This topological obstruction is exactly the nontrivial cohomology class.

9. Connection to MPS / SPT

For injective MPS:

$$U_g A^m = e^{i\phi(g)} V_g A^m V_g^{-1}$$

The projective class of V_g is classified by:

$$H^2(G, U(1))$$

In 1D bosonic systems:

$$\text{SPT phases} \leftrightarrow H^2(G, U(1))$$

Conceptual Flow

Associativity $\Rightarrow$ 2-cocycle

$\Rightarrow$ Gauge redundancy $\Rightarrow$ Cohomology group

$\Rightarrow$ Topological classification $\Rightarrow$ SPT invariant

Derived Stochastic Model from the AKLT State

We see that the AKLT state looks pretty much like a stochastic process which we have been investigating around, up to some phase freedom applied, we want to see the relation between the actual AKLT model and the stochastic version of it, and if there are any connections in symmetry.

The site matrices have the form:

$$\begin{bmatrix} 0. & 0.8165 \\ 0. & 0. \end{bmatrix}$$

$$\begin{bmatrix} 0.5774 & 0. \\ 0. & 0.5774 \end{bmatrix}$$

$$\begin{bmatrix} 0. & 0. \\ 0.8165 & 0. \end{bmatrix}$$

Calculating the Symmetry Group

Check if the ground space of the MPS changed:

$$\begin{bmatrix} 0. & 0. & 0. & 0. & 0. & 0.7698 & 0. & 0.6415 & 0. & 0. \\ 0. & 0.6415 & 0. & 1.0264 & 0. & 0.6415 & 0. & 0. & 0. & 0.6415 \\ 0. & 0.7698 & 0. & 0. & 0. & 0. & 0. & 0. & 0. & 0. \end{bmatrix}$$

As we can see, phase enhancing the MPS changes the ground space. We can find the symmetry group of this new MPS.

SYMMETRIC GENERATORS ANALYSIS

Found 1 generator(s) in the null space:

Generator 1:

Null space basis vector index: 0

Coefficients in local su(3) basis (orthonormalized): [0. 0. 0. 0. 0.
0. 0.5 0.866]

Basis generators in su(3) (with non-zero coefficients):

[6] (coeff: 0.5):

```
[[ 0.7071+0.j 0. +0.j 0. +0.j]
 [ 0. +0.j -0.7071+0.j 0. +0.j]
 [ 0. +0.j 0. +0.j 0. +0.j]]
```

[7] (coeff: 0.866):

```
[[ 0.4082+0.j 0. +0.j 0. +0.j]
 [ 0. +0.j 0.4082+0.j 0. +0.j]
 [ 0. +0.j 0. +0.j -0.8165+0.j]]
```

Local su(3) generator (Hermitian enforced):

```
[[ 0.7071+0.j 0. +0.j 0. +0.j]
 [ 0. +0.j 0. +0.j 0. +0.j]
 [ 0. +0.j 0. +0.j -0.7071+0.j]]
```

Orthonormalized null space basis (1 vectors):

Basis vector 1 (norm = 1.0000):

[0. 0. 0. 0. 0. 0. 0.5 0.866]

Virtual Symmetry

Physical generator G:

```
[[ 0.7071+0.j 0. +0.j 0. +0.j]
 [ 0. +0.j 0. +0.j 0. +0.j]
 [ 0. +0.j 0. +0.j -0.7071+0.j]]
```

Virtual generator found:

Phase: -0.000000-0.000000j

Error: 3.547699e-16

Relative error: 4.345026e-16

Generator K:

```
[[ -0.3536+0.j 0. +0.j]
 [ 0. +0.j 0.3536+0.j]]
```